KUTAISI INTERNATIONAL UNIVERSITY

Scientific Computing

Ramaz Botchorishvili

Kutaisi International University

1.6: Linear systems and the matrix exponential

Outline

Lecture 1.6: Linear systems with constant coefficients and the matrix exponential

- Eigenvalues and eigenvectors: distinct real, complex, repeated eigenvalues
- Matrix exponential and its computation by diagonalization
- Nonhomogeneous systems: variation of constants, undetermined coefficients, variation of parameters
- Variable coefficients: the exponential of the integral and its limits
- Q & A

Series: 1.1 classification. 1.2 models. 1.3 existence and uniqueness. 1.4 scalar closed-form solutions. 1.5 power series and Laplace. 1.6 linear systems.

Linear systems with constant coefficients

Definition 1.1 (Linear system with constant coefficients $\mathbf{x}'(t) = A\mathbf{x}(t)$ for $\mathbf{x} \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times n}$)

$\mathbf{x}' = A\mathbf{x}$ is the system $x_k' = \sum_j a_{kj} x_j$. For $n = 2$:

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

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Proposition 1.2 ($\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$ with $\mathbf{v} \neq \mathbf{0}$ solves $\mathbf{x}' = A\mathbf{x}$ if and only if $A\mathbf{v} = \lambda\mathbf{v}$)

Substituting: $\lambda\mathbf{v}e^{\lambda t} = A\mathbf{v}e^{\lambda t}$; divide by $e^{\lambda t}$. So λ is an eigenvalue ($\det(A - \lambda I) = 0$) and $\mathbf{v}$ an eigenvector ($(A - \lambda I)\mathbf{v} = \mathbf{0}$).

Distinct real eigenvalues

Theorem 1.3 (If A has distinct real eigenvalues λ_k with eigenvectors $\mathbf{v}_k$, the general solution of $\mathbf{x}' = A\mathbf{x}$ is $\sum_k c_k \mathbf{v}_k e^{\lambda_k t}$)

The n solutions $\mathbf{v}_k e^{\lambda_k t}$ are linearly independent (eigenvectors of distinct eigenvalues are), so

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{\lambda_1 t} + \cdots + c_n \mathbf{v}_n e^{\lambda_n t}.$$

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Example 1.4 ($A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$ has eigenvalues $3, -1$ and the general solution $c_1 \binom{1}{2} e^{3t} + c_2 \binom{1}{-2} e^{-t}$)

$\det(A - \lambda I) = (1-\lambda)^2 - 4 = 0$ gives $\lambda = 3, -1$. For $\lambda = 3$:
$(A - 3I)\mathbf{v} = \begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0}$, $\mathbf{v}_1 = \binom{1}{2}$. For $\lambda = -1$: $\mathbf{v}_2 = \binom{1}{-2}$.

Complex eigenvalues

Theorem 1.5 (A complex eigenpair $\lambda = \alpha + i\beta$, $\mathbf{v} = \mathbf{a} + i\mathbf{b}$ of a real A gives the real solutions $e^{\alpha t}(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t)$ and $e^{\alpha t}(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t)$)

$\mathbf{v}e^{\lambda t} = e^{\alpha t}(\mathbf{a} + i\mathbf{b})(\cos\beta t + i\sin\beta t)$ is a complex solution; its real part $\mathbf{x}_1$ and imaginary part $\mathbf{x}_2$ are real solutions:

$$\mathbf{x}_1 = e^{\alpha t}(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t), \qquad \mathbf{x}_2 = e^{\alpha t}(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t).$$

The general solution contributed by the pair is $c_1\mathbf{x}_1 + c_2\mathbf{x}_2$.

Complex eigenvalues: worked example

Example 1.6 ($A = \begin{pmatrix} -1 & 2 \\ -1 & -1 \end{pmatrix}$ has the eigenvalues $-1 \pm i\sqrt{2}$ and decaying oscillating solutions)

$\det(A - \lambda I) = (\lambda + 1)^2 + 2 = 0$, so $\lambda = -1 \pm i\sqrt{2}$, $\alpha = -1$, $\beta = \sqrt{2}$. For $\lambda = -1 + i\sqrt{2}$ the first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ gives $v_2 = \frac{i}{\sqrt{2}} v_1$:

$\mathbf{v} = \begin{pmatrix} \sqrt{2} \\ i \end{pmatrix} = \mathbf{a} + i\mathbf{b}$ with $\mathbf{a} = \begin{pmatrix} \sqrt{2} \\ 0 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

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$\mathbf{v} = \begin{pmatrix} \sqrt{2} \\ i \end{pmatrix} = \mathbf{a} + i\mathbf{b}$ with $\mathbf{a} = \begin{pmatrix} \sqrt{2} \\ 0 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

$$\mathbf{x} = c_1 e^{-t} \begin{pmatrix} \sqrt{2}\cos\sqrt{2}t \\ -\sin\sqrt{2}t \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} \sqrt{2}\sin\sqrt{2}t \\ \cos\sqrt{2}t \end{pmatrix}.$$

$\operatorname{Re}\lambda = -1 < 0$: the solutions oscillate and decay like e^{-t}.

Repeated eigenvalues and generalized eigenvectors

Proposition 1.7 (If $A\mathbf{v} = \lambda\mathbf{v}$ and $(A - \lambda I)\mathbf{w} = \mathbf{v}$, then $\mathbf{x}_1 = \mathbf{v}e^{\lambda t}$ and $\mathbf{x}_2 = (\mathbf{v}t + \mathbf{w})e^{\lambda t}$ solve $\mathbf{x}' = A\mathbf{x}$)

$\mathbf{x}_2' = \mathbf{v}e^{\lambda t} + \lambda(\mathbf{v}t + \mathbf{w})e^{\lambda t}$ *and* $A\mathbf{x}_2 = (\lambda\mathbf{v}t + A\mathbf{w})e^{\lambda t}$*; they agree because* $A\mathbf{w} = \lambda\mathbf{w} + \mathbf{v}$.

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Example 1.8 ($A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ has the solutions $e^{2t}\begin{pmatrix} c_1 + c_2 t \\ c_2 \end{pmatrix}$)

$\lambda = 2$ is double with the single eigenvector $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Solve $(A - 2I)\mathbf{w} = \mathbf{v}$: $\mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. So $\mathbf{x} = c_1\begin{pmatrix} 1 \\ 0 \end{pmatrix}e^{2t} + c_2\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}t + \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right)e^{2t}$.

Matrix exponential

Definition 1.9 (Matrix exponential $e^{At} = \sum_{k\geq 0}(At)^k/k!$)

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{(At)^k}{k!}.$$

The series converges for every square matrix A and every t.

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The series converges for every square matrix A and every t.

Theorem 1.10 (The IVP $\mathbf{x}' = A\mathbf{x}$, $\mathbf{x}(0) = \mathbf{x}_0$ has the unique solution $\mathbf{x}(t) = e^{At}\mathbf{x}_0$)

Differentiating the series term by term, $\frac{d}{dt}e^{At} = Ae^{At}$, and $e^{A\cdot 0} = I$. Uniqueness follows from lecture 1.3, since $A\mathbf{x}$ is Lipschitz.

Computing the matrix exponential by diagonalization

Proposition 1.11 (If $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_i)$, then $e^{At} = Pe^{Dt}P^{-1}$ with $e^{Dt} = \operatorname{diag}(e^{\lambda_i t})$)

$A^k = PD^kP^{-1}$, so the series gives $e^{At} = P\left(\sum_k \frac{(Dt)^k}{k!}\right)P^{-1} = Pe^{Dt}P^{-1}$. The columns of P are eigenvectors.

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Example 1.12 (For $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$ the IVP with $\mathbf{x}(0) = \binom{1}{0}$ has the solution $\mathbf{x} = \binom{(e^{3t}+e^{-t})/2}{e^{3t}-e^{-t}}$)

$$P = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix},\ D = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix},\ P^{-1} = \begin{pmatrix} 1/2 & 1/4 \\ 1/2 & -1/4 \end{pmatrix},$$

$$e^{At} = \begin{pmatrix} \frac{e^{3t}+e^{-t}}{2} & \frac{e^{3t}-e^{-t}}{4} \\ e^{3t}-e^{-t} & \frac{e^{3t}+e^{-t}}{2} \end{pmatrix}, \qquad \mathbf{x}(t) = e^{At}\begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Nonhomogeneous systems: variation of constants formula

Definition 1.13 (Nonhomogeneous linear system $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$)

A constant $n \times n$, $\mathbf{f}$ continuous. The general solution is $\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p$: general homogeneous solution plus one particular solution.

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Theorem 1.14 (The IVP $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$, $\mathbf{x}(0) = \mathbf{x}_0$ has the solution $\mathbf{x}(t) = e^{At}\mathbf{x}_0 + \int_0^t e^{A(t-\tau)}\mathbf{f}(\tau)\,d\tau$)

Differentiate: $\mathbf{x}' = Ae^{At}\mathbf{x}_0 + \mathbf{f}(t) + A\int_0^t e^{A(t-\tau)}\mathbf{f}(\tau)\,d\tau = A\mathbf{x} + \mathbf{f}$, *and* $\mathbf{x}(0) = \mathbf{x}_0$. *The first term solves the homogeneous system, the integral is a particular solution.*

Nonhomogeneous systems: undetermined coefficients and variation of parameters

Algorithm 1.15 (Undetermined coefficients for $\mathbf{x}' = A\mathbf{x} + \mathbf{f}$ with $\mathbf{f}$ built from polynomials, exponentials, sines and cosines)

1. Find $\mathbf{x}_h$.

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Algorithm 1.15 (Undetermined coefficients for $\mathbf{x}' = A\mathbf{x} + \mathbf{f}$ with $\mathbf{f}$ built from polynomials, exponentials, sines and cosines)

1. Find $\mathbf{x}_h$.
2. Guess $\mathbf{x}_p$ of the same type as $\mathbf{f}$ with unknown vector coefficients; e.g. for $\mathbf{f} = \begin{pmatrix} 2e^t \\ t \end{pmatrix}$ guess $\begin{pmatrix} a_1 e^t + b_1 t + c_1 \\ a_2 e^t + b_2 t + c_2 \end{pmatrix}$.

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3. If a term of the guess solves the homogeneous system, multiply it by t (or t^2).

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Proposition 1.16 (Variation of parameters for systems: $\mathbf{x}_p = \Phi(t) \int \Phi^{-1}(t)\mathbf{f}(t)\, dt$ for a fundamental matrix Φ of $\mathbf{x}' = A\mathbf{x}$)

Φ has linearly independent solutions as columns; the general solution is $\mathbf{x} = \Phi(t)\mathbf{c} + \mathbf{x}_p$.

Example: undetermined coefficients for a system

Example 1.17 ($\mathbf{x}' = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}\mathbf{x} + \begin{pmatrix} 0 \\ e^{4t} \end{pmatrix}$, $\mathbf{x}(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ has the solution $\mathbf{x} = -\frac{1}{2}\begin{pmatrix} 1 \\ 0 \end{pmatrix}e^{2t} + \begin{pmatrix} 1/2 \\ 1 \end{pmatrix}e^{4t}$)

Eigenvalues $2, 3$ with eigenvectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, so $\mathbf{x}_h = c_1\begin{pmatrix} 1 \\ 0 \end{pmatrix}e^{2t} + c_2\begin{pmatrix} 1 \\ 1 \end{pmatrix}e^{3t}$.

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4 is not an eigenvalue, so guess $\mathbf{x}_p = \begin{pmatrix} a \\ b \end{pmatrix}e^{4t}$: $4a = 2a + b$ and $4b = 3b + 1$ give $b = 1$, $a = \frac{1}{2}$.

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Initial condition: $\mathbf{x}(0) = \begin{pmatrix} c_1 + c_2 + 1/2 \\ c_2 + 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ gives $c_2 = 0$, $c_1 = -\frac{1}{2}$.

Example: variation of parameters for a system

Example 1.18 ($\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{x} + \begin{pmatrix} \sec t \\ 0 \end{pmatrix}$ has the particular solution $\mathbf{x}_p = \begin{pmatrix} t\cos t - \sin t \ln|\cos t| \\ -t\sin t - \cos t \ln|\cos t| \end{pmatrix}$)

$$\Phi = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} \text{ (columns } \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}, \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}\text{), } \Phi^{-1} = \Phi^T.$$

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$$\Phi^{-1}\mathbf{f} = \begin{pmatrix} 1 \\ \tan t \end{pmatrix}, \qquad \int \Phi^{-1}\mathbf{f}\, dt = \begin{pmatrix} t \\ -\ln|\cos t| \end{pmatrix}, \qquad \mathbf{x}_p = \Phi \begin{pmatrix} t \\ -\ln|\cos t| \end{pmatrix}.$$

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General solution: $\mathbf{x} = \Phi(t)\mathbf{c} + \mathbf{x}_p$.

Example: matrix exponential formula for a system

Example 1.19 ($\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}\mathbf{x} + \begin{pmatrix} e^t \\ 0 \end{pmatrix}$, $\mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ has the solution $\mathbf{x} = \begin{pmatrix} (3e^{3t}+e^{-t})/4 \\ \frac{3}{2}e^{3t} - e^t - \frac{1}{2}e^{-t} \end{pmatrix}$)

Use $\mathbf{x} = e^{At}\mathbf{x}_0 + \int_0^t e^{A(t-\tau)}\mathbf{f}(\tau)\,d\tau$ with e^{At} from the previous example. The first column of $e^{A(t-\tau)}$ times e^{τ} is integrated:

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$$\int_0^t \begin{pmatrix} \frac{e^{3(t-\tau)}+e^{-(t-\tau)}}{2} \\ e^{3(t-\tau)} - e^{-(t-\tau)} \end{pmatrix} e^{\tau}\,d\tau = \begin{pmatrix} (e^{3t} - e^{-t})/4 \\ (e^{3t} - 2e^t + e^{-t})/2 \end{pmatrix},$$

using $\int_0^t e^{-2\tau}\,d\tau = \frac{1-e^{-2t}}{2}$ and $\int_0^t e^{2\tau}\,d\tau = \frac{e^{2t}-1}{2}$.

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Use $\mathbf{x} = e^{At}\mathbf{x}_0 + \int_0^t e^{A(t-\tau)}\mathbf{f}(\tau)\,d\tau$ with e^{At} from the previous example. The first column of $e^{A(t-\tau)}$ times e^{τ} is integrated:

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using $\int_0^t e^{-2\tau}d\tau = \frac{1-e^{-2t}}{2}$ and $\int_0^t e^{2\tau}d\tau = \frac{e^{2t}-1}{2}$. Adding $e^{At}\mathbf{x}_0 = \begin{pmatrix} (e^{3t}+e^{-t})/2 \\ e^{3t}-e^{-t} \end{pmatrix}$ gives $x_1 = \frac{3e^{3t}+e^{-t}}{4}$ and $x_2 = \frac{3}{2}e^{3t} - e^t - \frac{1}{2}e^{-t}$.

Variable coefficients: the exponential of the integral

Proposition 1.20 (If $P(t)$ commutes with $\int_{t_0}^{t} P$ for all t, then $\Phi(t) = \exp\left(\int_{t_0}^{t} P(\tau)\, d\tau\right)$ is a fundamental matrix of $\mathbf{x}' = P(t)\mathbf{x}$)

Let $B(t) = \int_{t_0}^{t} P$. If $PB = BP$, then $\frac{d}{dt}B^k = kB^{k-1}P$, so differentiating the series of $\exp(B)$ term by term gives $\Phi' = P\Phi$. Then $\mathbf{x} = \Phi(t)\mathbf{c}$.

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Let $B(t) = \int_{t_0}^{t} P$. If $PB = BP$, then $\frac{d}{dt}B^k = kB^{k-1}P$, so differentiating the series of $\exp(B)$ term by term gives $\Phi' = P\Phi$. Then $\mathbf{x} = \Phi(t)\mathbf{c}$.

Example 1.21 (For $P(t) = \mathrm{diag}(t, 2t)$ the fundamental matrix is $\mathrm{diag}(e^{t^2/2}, e^{t^2})$)

$\int_0^t P = \mathrm{diag}(\frac{t^2}{2}, t^2)$ commutes with P (both are diagonal), so $\Phi = \mathrm{diag}\big(e^{t^2/2}, e^{t^2}\big)$ and $\mathbf{x} = \Phi\mathbf{c}$.

Without commutativity the exponential of the integral fails

Example 1.22 (For $P(t) = \begin{pmatrix} 0 & 1 \\ 0 & t \end{pmatrix}$ the matrix P does not commute with $\int_0^t P$ and $\exp(\int_0^t P)$ is not a solution)

$$B = \int_0^t P = \begin{pmatrix} 0 & t \\ 0 & t^2/2 \end{pmatrix}, \; PB = \begin{pmatrix} 0 & t^2/2 \\ 0 & t^3/2 \end{pmatrix} \neq BP = \begin{pmatrix} 0 & t^2 \\ 0 & t^3/2 \end{pmatrix}.$$

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$B = \int_0^t P = \begin{pmatrix} 0 & t \\ 0 & t^2/2 \end{pmatrix}$, $PB = \begin{pmatrix} 0 & t^2/2 \\ 0 & t^3/2 \end{pmatrix} \neq BP = \begin{pmatrix} 0 & t^2 \\ 0 & t^3/2 \end{pmatrix}$. With $\lambda = t^2/2$ we have $B^2 = \lambda B$, hence $\exp B = I + \frac{e^\lambda - 1}{\lambda} B$. The $(1,2)$ entry of $\Phi' - P\Phi$ is $\frac{t^2}{4} + O(t^4) \neq 0$.

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$B = \int_0^t P = \begin{pmatrix} 0 & t \\ 0 & t^2/2 \end{pmatrix}$, $PB = \begin{pmatrix} 0 & t^2/2 \\ 0 & t^3/2 \end{pmatrix} \neq BP = \begin{pmatrix} 0 & t^2 \\ 0 & t^3/2 \end{pmatrix}$. With $\lambda = t^2/2$ we have $B^2 = \lambda B$, hence $\exp B = I + \frac{e^\lambda - 1}{\lambda} B$. The $(1,2)$ entry of $\Phi' - P\Phi$ is $\frac{t^2}{4} + O(t^4) \neq 0$. So the commutation hypothesis cannot be dropped. In practice it is rarely satisfied, and $\exp(\int P)$ is of limited use for general variable coefficients.

References and further reading

- S. Pozza, N. Van Buggenhout, *A new matrix equation expression for the solution of non-autonomous linear systems of ODEs*, `arXiv:2210.07052` (2022).
- Boyce, DiPrima & Wilson, *Elementary Differential Equations and Boundary Value Problems*.
- All worked examples are verified symbolically by `closed_form_check.py`.

Q & A